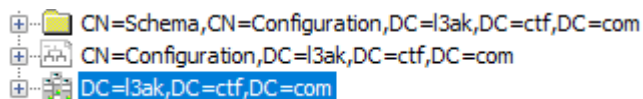
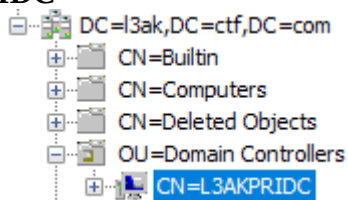


Once we discover that the Backup.dat file that is given is a backup of an Active Directory domain, we need to find a tool to analyze this file. This can be done with the tool AdExplorer.

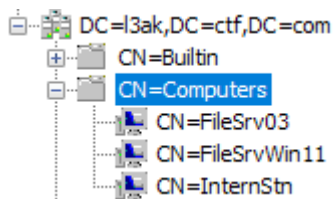
1) Once opened in AdExplorer, we can see multiple instances of "DN=l3ak,DC=ctf,DC=com" to give the answer **l3ak.ctf.com**.



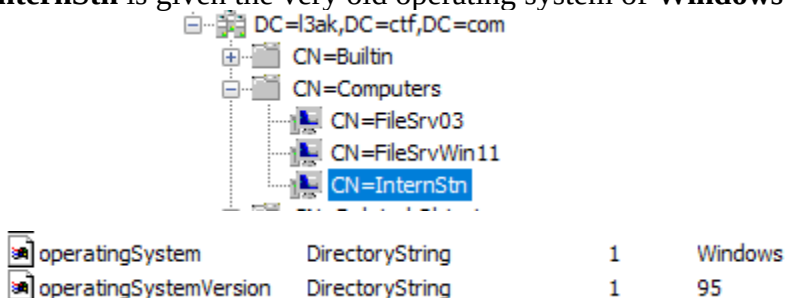
2) Expanding the field above, we can see an Organizational Unit (OU) called "Domain Controllers". This contains our answer: **L3AKPRIDC**



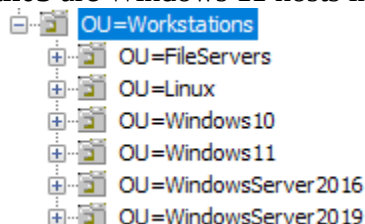
3) In Active Directory, hosts that have not been assigned to an OU, or ones that have been recently added to the domain, are in the "Computers" folder. There are 3 hosts that fit this criteria: **FileSrv03, FileSrvWin11, InternStn**



4) Computer objects in Active Directory contain two relevant attributes for this question: operatingSystem and operatingSystemVersion. After examining these values for all computer objects, we can see that the **InternStn** is given the very old operating system of **Windows 95**.

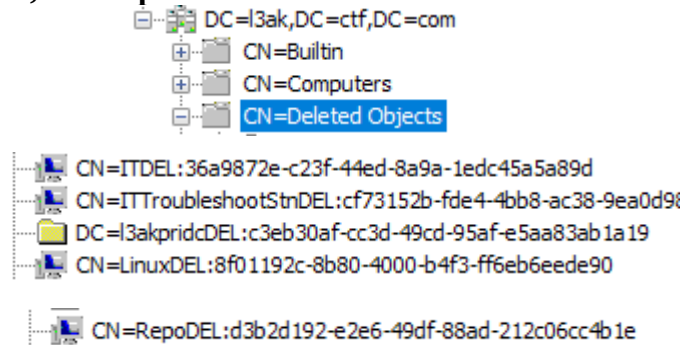


5) This involves further exploration of the computer objects in the domain, as well as the structure of the OUs in the domain. Under the "Workstations" OU we can see OUs for various operating systems. Both **ITWorkStn02** and **ITWorkStn03** are Windows 11 hosts in the Windows 10 OU.



 operatingSystem	DirectoryString	1	Windows
 operatingSystemVersion	DirectoryString	1	11

6) When an object is deleted in Active Directory, it is moved into the “Deleted Objects” container. Examining this container we can see the following computer objects have been deleted: **IT**, **ITTroubleshootStn**, **Linux**, and **Repo**.



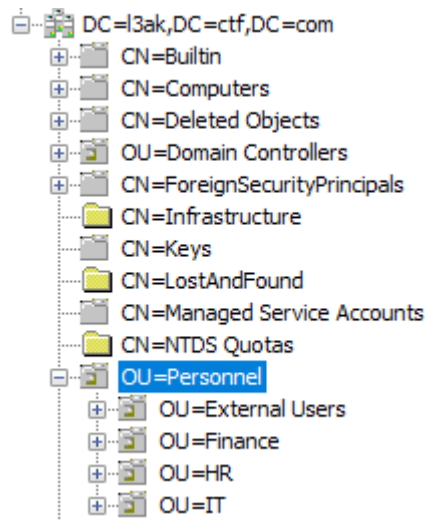
7) In Active Directory, the userAccountControl variable gives information about the status of a user account. These values are described here: <https://learn.microsoft.com/en-us/troubleshoot/windows-server/active-directory/useraccountcontrol-manipulate-account-properties> and <https://jackstromberg.com/2013/01/useraccountcontrol-attribute-flag-values/>. For a disabled account, we need a hex value of **0x202**. AdExplorer shows values in decimal by default, but we can right click on an attribute, select “Display Integers as”, then choose hex. Now we just need to find all users who have a userAccountControl attribute of 0x202. This corresponds to the following users: **Wilhelm Firtz**, **Reginald Norwood**, and **Christopher Price**.

 userAccountControl	Integer	1	0x202
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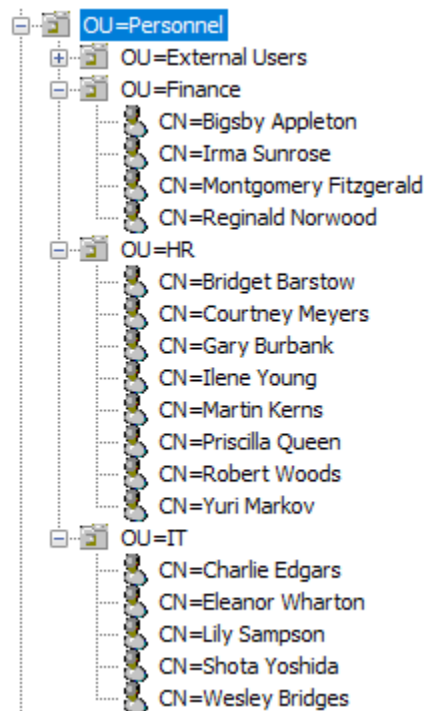
8) Using the references above, we can see that we need to find a userAccountControl value of **0x10200**. This corresponds to the following users: **Bigby Appleton**, **Montgomery Fitzgerald**, and **Lily Sampson**.

 userAccountControl	Integer	1	0x10200
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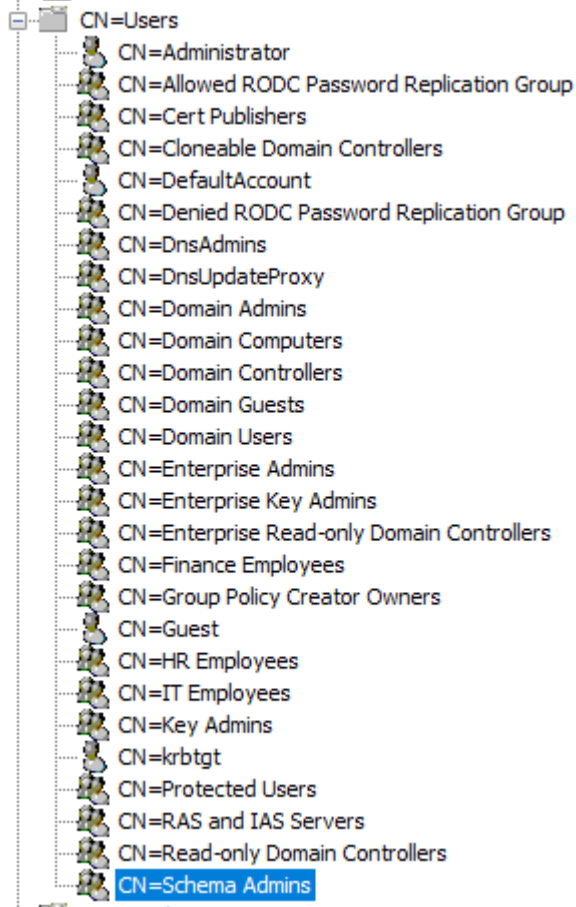
9) We can see an OU called “Personnel”, and this is where we will find our various departments as well as the ability to count the number of employees in each department. The departments are **Finance**, **HR**, and **IT**, as shown below:



Expanding each OU we can count the number of user accounts in each OU. However, the question asks for *active* employees. Reginald Norwood in the Finance OU has their account set to Disabled – which is shown in the userAccountControl attribute being 0x202. Therefore, the final answer is as follows:
Finance-3, HR-8, IT-5



10) In an Active Directory forest, the users who have the most control over its structure are those in the “Schema Admins” group. Under the Users folder, clicking on the Schema Admins group we can see in the “member” attribute that **Charlie Edgars** and **Lily Sampson** are the only members of this group.



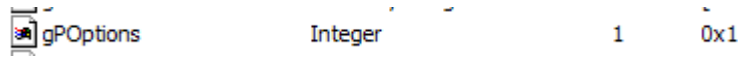
	member	DN	3	CN=Charlie Edgars,OU=IT,OU=Personnel,DC=l3ak,DC=ctf,DC=com;CN=Lily Sampson,
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11) The principle of least privileges states that users should, at the onset of their account creation, be given access to nothing and then only the necessary permissions are given to them. These permissions can be based on many things – geographic location, user role in the company, or department the user belongs to to name a few. For this domain, we can see that both **Christopher Price** and **Eleanor Wharton** are members of groups outside of their OU.

	memberOf	DN	4	CN=IT Employees,CN=Users,DC=l3ak,DC=ctf,DC=com;CN=Finance Employees,
	name	DirectoryString	1	Eleanor Wharton
	memberOf	DN	1	CN=IT Employees,CN=Users,DC=l3ak,DC=ctf,DC=com
	name	DirectoryString	1	Christopher Price

12) The attribute that defines if inheritance is blocked or not for an OU is the gPOptions attribute. Further research on this attribute can be found here:

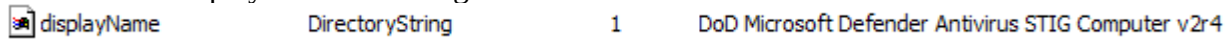
https://learn.microsoft.com/en-us/openspecs/windows_protocols/ms-gpol/08090b22-bc16-49f4-8e10-f27a8fb16d18. We can see that the following OUs have this attribute set to 0x1: **Domain Controllers**, **IT**, and **FileServers**.



gPOptions	Integer	1	0x1
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13) To view the Group Policy Objects (GPOs) for a domain, we need to expand the “System” folder, and then the “Policies” folder under that. Here, we can see the displayName attributes for these GPOs start with “DoD”. This stands for Department of Defense. If we google the phrase “united states dod gpos” we are taken to the following link: [https://public.cyber.mil/stigs/gpo/#:~:text=%E2%80%8BGroup%20Policy%20Objects%20\(GPOs,group%20of%20computers%20or%20users](https://public.cyber.mil/stigs/gpo/#:~:text=%E2%80%8BGroup%20Policy%20Objects%20(GPOs,group%20of%20computers%20or%20users). Here we can download the file called “Group Policy Objects (GPOs) – April 2025” and take the sha256sum, equating to **4BD7742C73A610EDF79A6B484457351438C90DC6FAC119EF8475B46D96BD2B37**.

14) This requires some further OSINT based on analysis of the GPOs found above. One of the GPOs we can see has a displayName including **Microsoft Defender**. This is our first answer.



displayName	DirectoryString	1	DoD Microsoft Defender Antivirus STIG Computer v2r4
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Googling the entire value for the attribute leads us to the following website:

https://www.tenable.com/audits/DISA_STIG_Microsoft_Defender_Antivirus_v2r4. Here we can find the following two rules: *WNDF-AV-000029 - Microsoft Defender AV virus definition age must not exceed 7 days* and *WNDF-AV-000036 - Microsoft Defender AV must be configured to impede JavaScript and VBScript to launch executable*. These give us our other answers of **7 days** and **JavaScript** and **VBScript** being impeded from launching executables.